UNIVERSITÀ DEGLI STUDI DI MILANO BICOCCA

Foundations of Deep Learning

Flower Image Classification

Nicole Winy Hernandez 873125
Azriel Matthew Ortega 946814
Meryann Joy Tibayan 886764

Introduction

The **task** of this project is to **correctly classify** the images belonging to the dataset into 5 specific flower category: tulips, dandelion, daisy, sunflowers and roses

Challenges:

- the flowers share very similar backgrounds and structures, requiring models to focus on fine textures and specific patterns.
- Risk of Overfitting: With a small dataset (3670 images), neural networks tend to "memorize" the training photos, struggling to generalize to new images.

Objective:

What level of architectural complexity is truly required to solve this? Can a purpose-built CNN trained from scratch cut it, or do modern pre-trained backbones completely rewrite the rules of efficiency and accuracy?

Data Preparations

Data Cleaning

Removed images that:

- dont covert atleast 20% of the image
- Not the main object of focus
- two or more flowers
- Zoomed out

Why?
To focus on the important aspects of the flower such as the shape, colors etc.







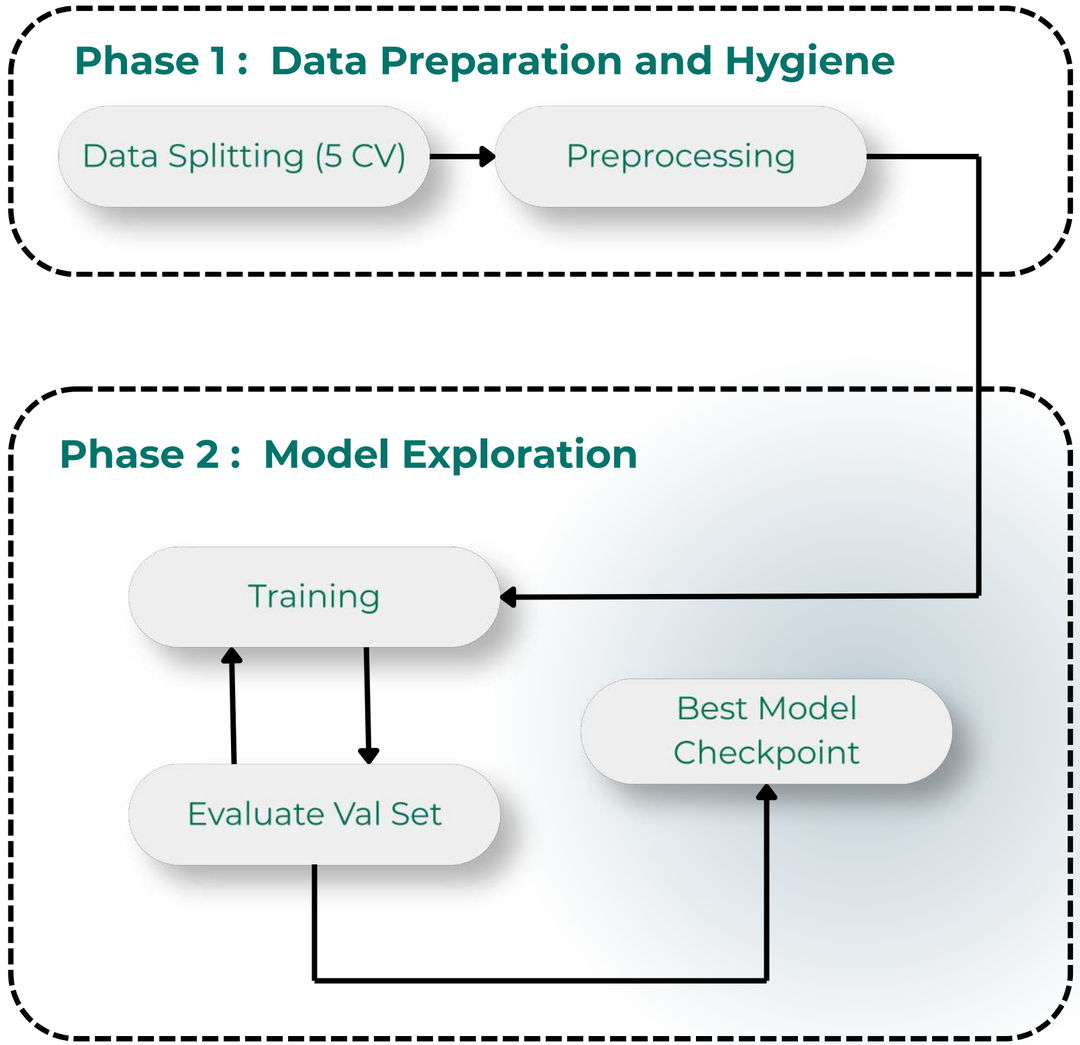
Data Preprocessing

- **Resize Image**s: 224x224
- **Normalizations**:
  - [-1, 1] for ScratchCNN and MobileNet
  - ViT does normalization during training
- **Augmentation** MobileNet and ViT: flip, rotation, zoom, contrast, brightnes

Training Setup & Data splitting

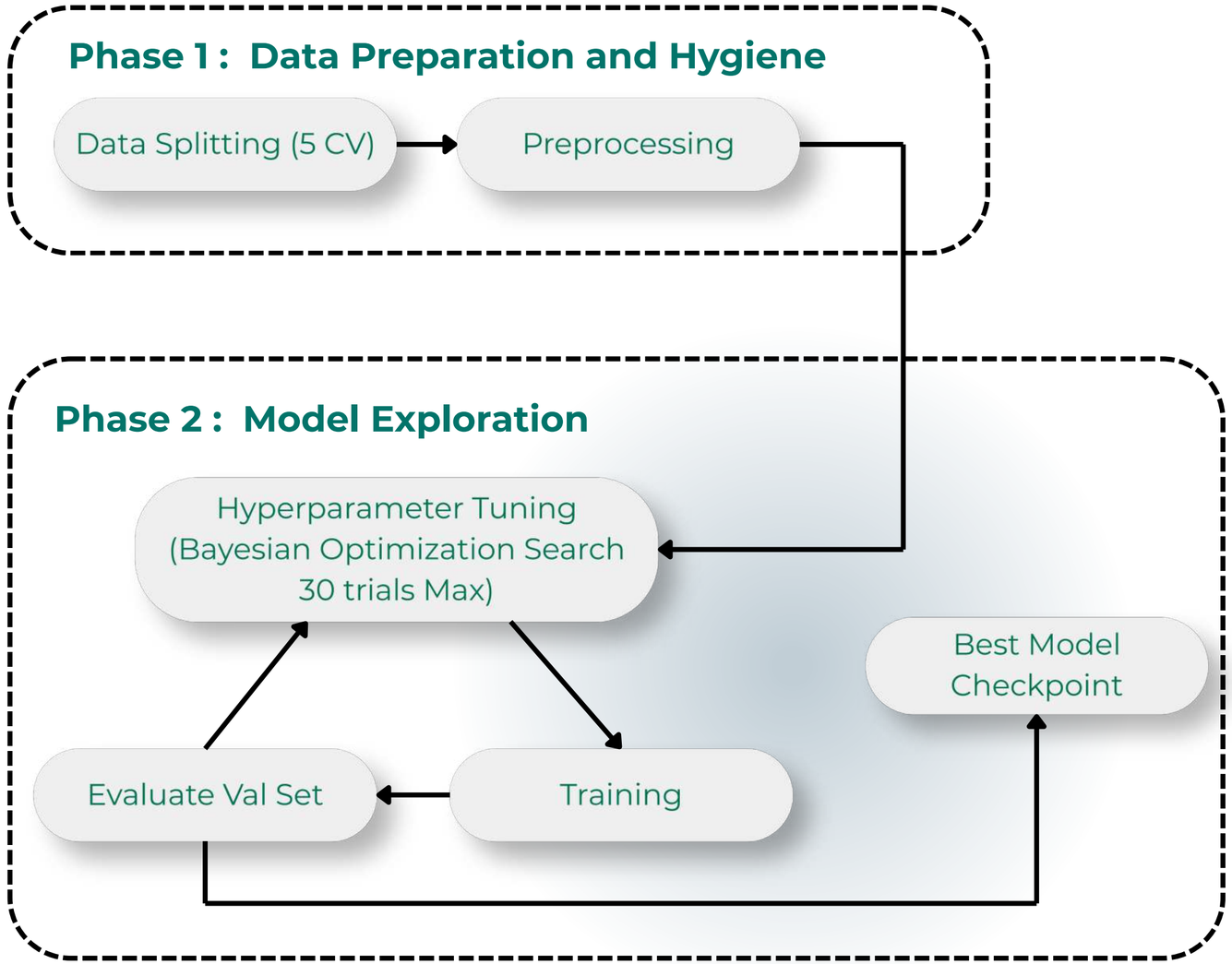
- **Batch size**: 32
- **Optimizer**: Adam
- **Loss Function**: Sparse Categorical Crossentropy*
- **Initial Split:** 80% train, 20% test
- **5-Fold Cross-Validation:**
  - ~80% training (~2,253–2,254 images)
  - ~20% validation (~563–564 images) per fold

ScratchCNN - v1



Paradigm	Standard 2D Convolutions
Pre-training	None
No. of Parameters	Total params: 101,829 Trainable params: 101,829 Non-trainable params: 0
Input Shape	224,224,3
Optimizer & Loss	Adam (Initial LR: $10^{-3)}$ Sparse Categorical Crossentropy
Backbones	Conv Blocks [32, 64, 128] filters (3x3) Convolutions + MaxPool (2x2)
Regularization	Dropout (0.2), L2 Penalty (1e-4), Batch Normalization
Training	5 - fold CV Early Stopping (patience= 7)

ScratchCNN - Final



Paradigm	Standard 2D Convolutions
Pre-training	None
No. of Parameters	Total params: 1,687,237 Trainable params: 1.684,805 Non-trainable params: 2,432
Classification Head	GAP → Dropout(0.2) → Dense(256, L2) + BatchNorm + ReLU → Dropout(0.2) → Dense(5, Softmax)
Input Shape	224,224,3
Optimizer & Loss	Adam (Initial LR: $10^{-3)}$ + LR Scheduler / Decay Sparse Categorical Crossentropy
Backbones	Conv Blocks [64, 128, 256, 512] filters (3x3) Convolutions (same padding) BatchNorm + ReLU + MaxPool (2x2)
Regularization	Dropout (0.2), L2 Penalty (1e-4), Batch Normalization
Training	5 - fold CV LR Scheduler + Early Stopping (patience= 7)

ScratchCNN - Experiments

Milestone #	Focus Area	Modifications	Result
M1	**Achitecture Search** Find optimal baseline capacity	• Establish custom architecture • Bayesian Search Tuning (30 trials) • Data Augmentation Experiment (Failed)	Val Accuracy: 0.75 Overfitting
M2	**Input Hygiene** Stabilize Gradients	• [-1, 1] normalization • Resize logic to add pad	Val Accuracy: 0.78
M3	**Dataset & Overfitting** Sanitize Inputs & Regularize	• Batch: 16 → 32 • Removed bad images • Dropout: 0.2 added + Tuned Learning Rate	Val Accuracy : 0.78 Lessen Generalization Gap Decrease Optimization Stability
M4	**Imbalance & Convergence** Find convergence tuning	• Resizing logic to resize only • Added ReduceLROnPLateau • Class Weights added	Val Accuracy : 0.84

MobileNetV2

MobileNetV2 is a CNN pre-trained on ImageNet. Depthwise Separable Convolutions
The original classifier is removed and the backbone is reused as a feature extractor.

Classification Heads

The MobileNetV2 backbone is kept frozen.
Three feed-forward classification heads:
1. simple: no hidden Dense layer
2. medium: one hidden Dense layer
3. heavy: two hidden Dense layers

Does increasing head complexity lead to better performance?

Light fine-tuning

Only the last 20 MobileNetV2 layers are **unfrozen**.

Does light fine-tuning improve MobileNetV2 performance on flower classification?

Vision Transformer - ViT

ViT is a patch-based architecture, pre-trained on ImageNet-21k, that splits an image into 196 patches and uses **multi-head self-attention** to model global relationships between them from the very first layer.

Feature extraction

Setup:

- backbone frozen (85.8M weights),
- only the head (3,845 params) trained at lr=1e-4

What can ViT achieve with zero backbone adaptation?

Two-Stage Fine-Tuning

Setup:

- **Stage 1:** epochs 1-15 warm up the head at lr=1e-4, backbone frozen;
- **Stage 2**: at epoch 15 the backbone unfreezes and lr drops to 1e-5 with clipnorm=1.0

What happens if we unfreeze all ~86M parameters?

Model Selection & Validation Results

Model	Trainable Params	Non Trainable Params	Mean Val Accuracy	Mean Val Macro F1
Scratch	1.685,237	2,432	0.8346	0.8316
MobileNetV2 Simple Head	6,405	2,257,984	0.9070	0.9053
MobileNetV2 Medium Head	164,613	2,257,984	0.9052	0.9039
MobileNetV2 Heavy Head	361,477	2,257,984	0.9049	0.9031
MobileNetV2 Simple Head Fine Tuned	1,201,285	1,063,104	0.9152	0.9136
ViT	3,845	85,798,656	0.9677	0.9667
ViT Fine Tuned	85,802,501	0	**0.9822**	**0.9815**

Pre-training delivered the **largest gain**: all MobileNetV2 variants clearly outperformed Scratch. Increasing head complexity led to only marginal differences, while light fine-tuning produced a modest additional gain

Fine-tuned ViT achieved the highest score, but **frozen ViT** delivered similar performance while training far fewer parameters.

Test Set Results

Model Family	Mean Fold Test F1	Ensemble Test F1	Stability (F1 Spread)	Speed (ms)	Takeaway
ViT	~0.9675	0.9719	0.0056 (Extremely tight)	211.91	Dominates all metrics; benefits from ensembling.
MobileNetV2	~0.8887	0.8938	0.0141 (Stable)	146.12	Strong accuracy (~89.5%) with low parameter cost.
SCRATCH	~0.8146	0.8245	0.0195 (Acceptable)	16.98	Solid custom baseline; clearly proves the value of pretrained representations.

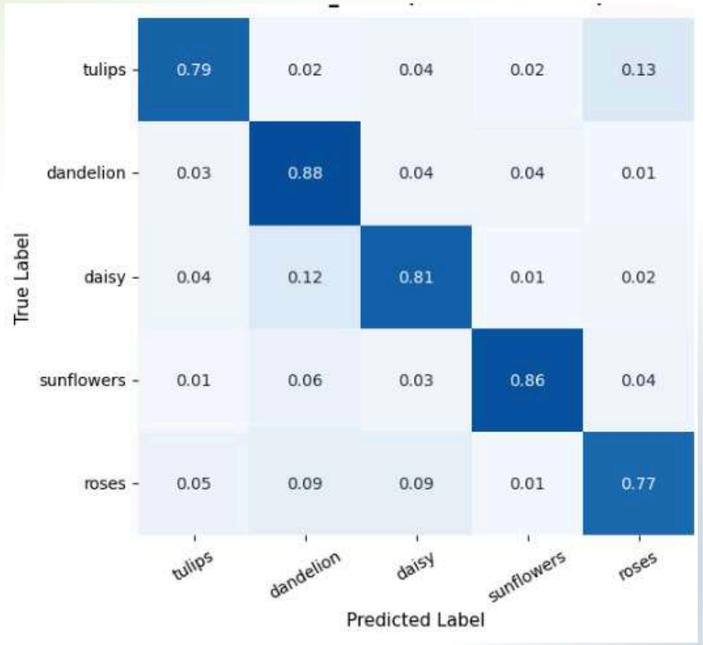
Key Takeaways

- ViT Dominates: Achieves top F1 score (0.972) with near-zero variance across folds.
- Pretraining Closes the Gap: Drives massive gains on hard classes (Roses +18pp, Tulips +17pp).
- Speed vs. Accuracy: Scratch is fastest (17 ms), but ViT delivers production-grade reliability.

Class	Support	SCRATCH (F1)	MobileNetV2 (F1)	ViT (F1)
Daisy	160	0.8301	0.8972	0.9592
Dandelion	180	0.8427	**0.9101**	0.9889
Roses	126	0.7969	0.8943	0.9802
Sunflowers	139	**0.8856**	0.8889	**0.9891**
Tulips	128	0.7674	0.8784	0.9421

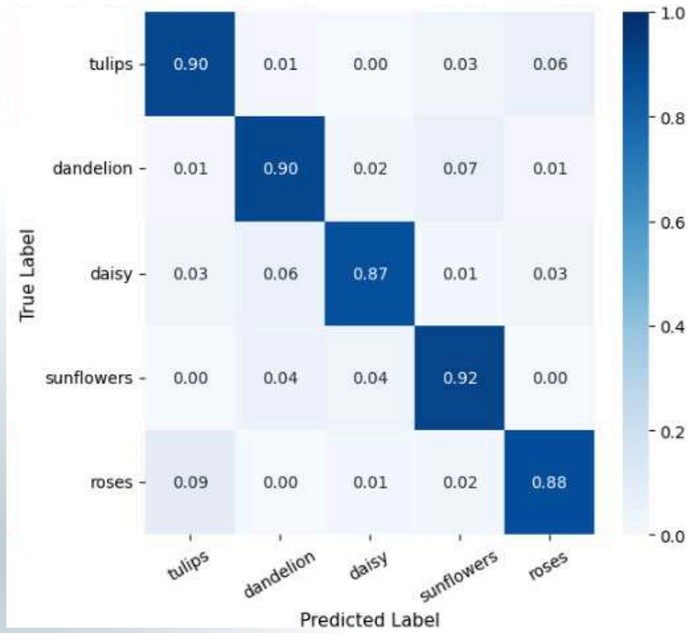
Test Set Results

SCRATCH CNN



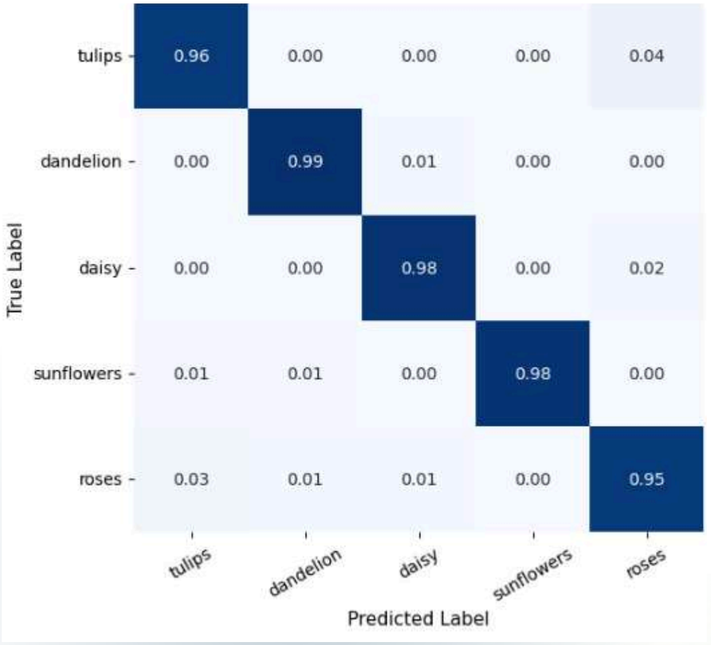
- F1: 0.825 | High Confusion
- Frequent leakage across classes
- Daisy are often misclassified as dandelion and tulips as roses

MobileNetV2 Fine Tuned



- F1: 0.894 | Suppressed Errors
- Resolves color bleed, residual error concentrated in silhouette overlap

ViT



- F1: 0.972 | Near-Perfect Diagonal
- Diagonal reaches 95%–99%; cross-class confusion virtually eliminated.

Error Analysis

SCRATCH CNN



Predicted : Tulips
True : Roses
Color Shortcut Bias



Predicted : Daisy
True : Sunflowers
Background Foliage



Predicted : Roses
True : Tulips
Petal Geometry Confusion

MobileNetV2



Predicted : Tulips
True : Roses
Morph Ambiguity



Predicted : Sunflowers
True : Roses
Shadows / Low Contrast



Predicted : Dandelion
True : Sunflowers
Background Blur

ViT



Predicted : Tulips
True : Roses
Morph Ambiguity



Predicted : Roses
True : Daisy
Multi Class



Predicted : Dandelion
True : Tulips
Dataset Anomaly

CONCLUSIONS

WHAT WE LEARNED AND MODEL LIMITATIONS

1. Scratch CNN

Is sufficient as a lightweight and fast baseline. However, it does not learn features robust enough to generalize well from a small dataset. The model is more affected by overfitting and by images with cluttered backgrounds, unusual viewpoints, or visually similar flowers

3. Fine-Tuning

Fine-tuning improved both MobileNetV2 and ViT, confirming that adapting pre-trained features to the target dataset is useful. However it is important to consider whether the additional accuracy justifies the greater training time, memory use, and computational cost.

2. Pre-training Advantage

MobileNetV2

Starting from pre-trained features gives a huge improvement over the Scratch CNN. While making the final classification head deeper did not bring a meaningful gain: better features mattered more than a larger head.

ViT

ViT achieved the strongest and most stable results. Its pre-trained representation was better able to distinguish subtle visual differences between flower classes.

EFFICIENCY TRADE-OFF

4. Efficiency Trade-off

Higher accuracy, higher cost:
Pre-trained models achieve better classification results because they start from stronger visual features, but they are larger and take longer to process each image than the custom CNN.

While all models provide fast predictions for a single image in our benchmark, the Scratch CNN is more suitable for real-time or high-throughput use cases.
MobileNetV2 and ViT are preferable when classification accuracy is the main priority.

Thank you